

Smart Attendance System: Complete Deployment & Cost Quotation Report

1. Executive Summary & Infrastructure Overview

This document provides a comprehensive cost estimation, resource allocation model, and deployment strategy for launching the **Smart Attendance System** across Cloud Infrastructure, Web, Android, and iOS platforms.

The application utilizes a modern, decoupled microservice architecture:

- **Frontend Layer:** React.js (v19) + Vite, wrapped with Ionic Capacitor for Android & iOS native execution.
- **Core API Layer:** Node.js (v20) + Express.js RESTful API engine handling session management, business logic, and database operations.
- **Biometric Engine:** Python FastAPI microservice utilizing the **InsightFace** `buffalo_sc` (**Small Footprint**) neural network model for facial detection and 512-dimensional vector embedding comparison.
- **Database Layer:** MongoDB document-oriented cloud cluster managed via Mongoose ODM.

2. Itemized Deployment Cost Matrix

Below is the complete financial breakdown categorized by component, cloud vendor, billing frequency, and cost in USD.

Category	Provider / Service	Billing Model	Cost (USD)	Service Capacity / Limits
Frontend Web Hosting	Vercel / Netlify	Free Tier	\$0.00 / mo	Unlimited bandwidth, global CDN edge caching, SSL included
Database	MongoDB Atlas (M0)	Free Tier	\$0.00 / mo	512 MB Storage, ~500,000 attendance records,

	Cluster)			automated backups
Core API & Microservice	DigitalOcean (Basic Droplet)	Monthly	\$4.00 - \$6.00 / mo	1 vCPU, 512MB - 1GB RAM, 25GB NVMe SSD, 1TB Transfer
Domain Name	Namecheap / Cloudflare	Annual	\$10.00 - \$12.00 / yr	Required for SSL certificate & secure API routing
SSL Security Certificate	Let's Encrypt / Certbot	Permanent	\$0.00	Automated TLS 1.3 encryption for HTTPS compliance
Push Notifications	Firebase Cloud Messaging (FCM)	Free Tier	\$0.00	Unlimited push alerts to Android & iOS mobile devices
Android Publishing	Google Play Console	One-Time	\$25.00	Lifetime developer account to publish unlimited apps
iOS Publishing	Apple Developer Program	Annual	\$99.00 / yr	Annual membership to publish on Apple App Store & TestFlight

3. Detailed Breakdown of Free Tiers & Performance Limits

Many core components of this system can be hosted permanently on enterprise-grade free tiers without sacrificing speed, security, or reliability.

A. Frontend Static Web Hosting (Vercel / Netlify)

- **Why it is \$0:** Once compiled via `npm run build`, the React frontend consists entirely of static HTML, JavaScript, and CSS assets. Static files do not require server CPU compute time.
- **Stats & Specifications:**
 - **Bandwidth:** 100 GB / month (free).
 - **Deployments:** Continuous Integration (CI/CD) automatically deploys every GitHub commit.

- **Speed:** Assets are cached on Global Edge Networks (CDN) close to the user, resulting in sub-50ms page load times.

B. Cloud Database (MongoDB Atlas M0 Cluster)

- **Why it is \$0:** MongoDB Atlas provides a permanent free-tier M0 cluster for lightweight applications.
- **Stats & Specifications:**
 - **Storage:** 512 MB.
 - **Record Capacity:** An attendance log entry averages ~1 KB in size. 512 MB can comfortably store over **500,000 attendance records** before requiring a storage upgrade.
 - **Connections:** Supports up to 500 simultaneous connections.

C. Push Notification Engine (Firebase FCM)

- **Why it is \$0:** Google provides Firebase Cloud Messaging (FCM) free of charge to all mobile developers.
 - **Stats & Specifications:**
 - **Volume:** Unlimited push notifications per day.
 - **Reliability:** 99.9% delivery rate across both Android and iOS background processes.
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4. DigitalOcean Server Configuration & Memory Management

The Core Backend (Node.js) and the AI Microservice (Python `buffalo_sc`) will run side-by-side on a single DigitalOcean Ubuntu Linux Droplet.

Server Specifications

- **Operating System:** Ubuntu 22.04 LTS x64
- **Instance Type:** Basic Droplet (\$4.00/mo for 512MB RAM or \$6.00/mo for 1GB RAM)
- **CPU:** 1 vCPU

Memory Budget Breakdown:

- **Ubuntu OS Baseline:** ~120 MB RAM
- **Node.js Express Server:** ~90 MB RAM
- **Python InsightFace (`buffalo_sc`):** ~250 MB RAM

- **Total Operational Memory: ~460 MB RAM**

Mandatory Optimization: The 1GB Swap File Configuration

To prevent out-of-memory (OOM) crashes on a 512MB server during peak morning check-in spikes or `npm` installations, a **1GB Swap Space** must be configured on Linux. Swap Space allows the operating system to use SSD storage as virtual RAM when memory usage approaches 100%.

Linux Commands for Swap Configuration:

```
sudo fallocate -l 1G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
```

5. Mobile App Publishing Costs & Platform Requirements

Deploying the React application to mobile devices using **Ionic Capacitor** involves platform-specific developer requirements.

A. Android Platform (Google Play Store)

- **One-time Fee:** \$25.00 USD (Lifetime access)
- **Compilation Requirements:** Android Studio running on any Windows, macOS, or Linux computer (Free tool).
- **Output Artifact:** `.aab` (Android App Bundle) or `.apk`.

B. iOS Platform (Apple App Store)

- **Annual Fee:** \$99.00 USD / year (Required to keep the app live)
- **Hardware Requirement: macOS Computer (Mandatory)**
 - *Note: Apple strictly requires Xcode to build and sign iOS applications. Xcode only runs on macOS.*
 - *If a physical Mac computer is unavailable, a cloud-based Mac instance (e.g., MacInCloud) can be rented for ~\$20/month during build cycles.*

6. Comprehensive Budget Scenarios

Scenario A: Ultra-Lean / Minimum Cost Setup

Designed for small teams, startups, or internal testing.

- **Initial Day 1 Expenses:** \$25.00 (*Google Play Console*)
- **Monthly Operational Cost:** \$4.00 / month (*DigitalOcean 512MB Droplet*)
- **Annual Recurring Cost:** \$10.00 / year (*Domain Name*)
- **Year 1 Total Cost: \$83.00 USD** (*Excludes Apple App Store*)

Scenario B: Full Commercial Production (Android + iOS + Web)

Designed for complete corporate deployment across all platforms.

- **Initial Day 1 Expenses:** \$25.00 (*Google Play Console*)
 - **Annual Subscriptions:** \$99.00 (*Apple Developer*) + \$12.00 (*Domain*) = **\$111.00 / year**
 - **Monthly Server Cost:** \$6.00 / month (*DigitalOcean 1GB Droplet*) = **\$72.00 / year**
 - **Year 1 Total Cost: \$208.00 USD**
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7. App Store Review & Compliance Checklist

To ensure approval during Google and Apple review processes, the following items must be implemented:

1. Native Permissions Strings:

- **Camera:** `NSCameraUsageDescription` added to `Info.plist` (Required for face scan).
- **Geolocation:** `NSLocationWhenInUseUsageDescription` added to `Info.plist` (Required for geofencing).

2. HTTPS Mandate:

All production API endpoints must use SSL encryption (`https://api.yourdomain.com`). Plain `http://` calls are blocked by Android 9+ and iOS natively.

3. Public Privacy Policy:

A published Privacy Policy link explicitly explaining how facial vector data and GPS coordinates are stored and encrypted.